

PERSONAL STATEMENT

My interest in computer science originated from my fascination with the underlying working principles of systems. Starting from independently seeking resources on the internet in the early days, to assembling computers, debugging hardware and operating systems by myself, I made the transition from a passive user to an active explorer. This also convinced me of the importance of a solid theoretical foundation in computer science for technological innovation. Moreover, nowadays, although the breakthroughs in fields such as artificial intelligence inspire me, I still realize more deeply that what lies behind them is efficient algorithms, powerful computing power and reliable systems. Enlightened by this, I am eager to participate in the MSc Computer Science program at the City University of Hong Kong. I intend to systematically deepen my knowledge in core areas such as algorithms and distributed systems, evolving from a developer who can implement functions to an engineer who can contribute to building the next generation of reliable and efficient computing systems.

My undergraduate studies has laid a solid theoretical and engineering foundation for me to enter the field of computer science. In the course *Algorithm Design and Analysis*, I not only mastered common algorithm paradigms and complexity analysis skills, but also developed sensitivity to computational efficiency issues. In addition, the course *Introduction to Software Engineering* has enabled me to initially establish a systematic understanding of system architecture design and standardized development processes. I place particular emphasis on applying theory to practice. In the team project for developing a traffic query system, I took the initiative to undertake the development of the front-end module. By deeply studying the reactive principle and component-based design of Vue.js, I independently completed the modular interface design and data interaction logic. This experience has made me deeply understand that a solid theoretical foundation is the cornerstone for quickly mastering new technologies. During this process, aiming at systematizing the scattered experiences, I established a personal GitHub knowledge base, continuously archiving project code, technical solutions and learning notes. This action has prompted me to constantly refactor and optimize existing codes, gradually forming a set of reusable and iterative personal development system. Through this continuous practice and reflection, I clearly realized that to evolve from a developer who implements specific functions to an engineer capable of designing complex systems, one must conduct deeper exploration in core areas of computer science such as algorithm optimization, distributed systems and artificial intelligence.

Realizing this, I have been keeping with the passion for investigating computer science in a great scenario. When undertaking one-month server operation internship, I directly encountered the gap between theoretical plans and engineering reality, enabling me to connect theoretical knowledge with actual engineering constraints. In a data migration task, the initial verification script based on static configuration failed due to the dynamic updates of the backup. I quickly adjusted the plan and solved the problem by calling APIs to dynamically obtain the server status. This experience deepened my understanding of the robust system: it must strike a balance between theoretical correctness and the dynamic complexity of the actual environment. More importantly, during an automatic backup task of the bastion host, I accidentally deleted the configuration files due to excessive permissions. Under the guidance of a senior engineer, I learned to locate the fault through segmented logs and established a systematic troubleshooting mindset. This lesson has given me a profound understanding of the security boundaries and system reliability of the production environment, and has also cultivated in me the necessary rigor and calmness when dealing with complex system issues.

The challenges encountered at the engineering front line have made me clearly realize that scattered practices cannot replace a systematic theoretical foundation. Now, I not only aim to fix problems but also strive to build a next-generation software architecture with inherent safety and high reliability. To achieve this, I must master its core principles. I firmly believe that the MSc Computer Science program at City University of Hong Kong is the decisive step towards this goal. I look forward to thoroughly systematizing my practical experience through its top courses in distributed systems, advanced algorithms, and information security, thereby acquiring the key ability to design complex systems rather than merely fix code.

Applicant: LUO Xiaoyu
Program: MSc Computer Science, City University of Hong Kong

I have learned that this program offers a solid theoretical foundation, and possesses significant technical advantages: its curriculum setting is closely aligned with the most cutting-edge cloud computing architectures and distributed system principles in the industry. With these advantages, I will be able to deeply understand the underlying engineering implementation of machine learning systems through practical experience. I am convinced that this is exactly what I am striving for: to elevate my development skills to a system-level capability that can handle and optimize complex software development cycles. After carefully studying the curriculum system of your program, I plan to gain a deep understanding of the underlying principles of data routing, transmission control, and the network protocol stack through the course *Computer Networks and the Internet*. This will serve as the theoretical foundation for me to optimize the communication efficiency of distributed systems and address the performance bottlenecks at the network layer in the future. At the same time, the Software Engineering course will guide me to develop a systematic architectural design mindset and learn to implement strict code quality control and efficient team collaboration processes in large and complex projects.

With the expertise included in your program, I am looking forward to deeply integrating and applying the knowledge in practice. During my studies, I plan to use the course projects as the core testing ground: applying the modularization and high cohesion principles of software engineering to design a microservice architecture. Based on this, I plan to leverage your program's intelligent system laboratory and cloud computing platform to build a self-adaptive distributed system in the course project. Through continuous monitoring and dynamic scheduling, I aim to optimize the utilization of resources and achieve the communication goal of low latency and high throughput. I believe this complete closed loop from abstract theory to specific implementation, followed by feedback through system performance indicators, is precisely the key step for me to convert cross-domain knowledge into deliverable and verifiable system-level solutions.

With the support of your program, my long-term career vision is to become a research and development engineer who specializes in building highly reliable and scalable software infrastructure. I aspire to be involved in and even lead the development of industry-level cloud-native platforms, directly facing and addressing core challenges such as service guarantee and seamless fault switching under the load of massive data throughput. I firmly believe that the systematic training I will receive at CityU will not only endow me with the ability to build functions but also provide me with the vision and authority to define and validate the foundation of the future digital world.



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Recommendation Letter

Date: September 23, 2025

To whom It May Concern,

It is my pleasure to write this recommendation letter for LUO Xiaoyu, one of my best students in the course *Introduction to Computer Science and Technology*, to support him in applying for this esteemed postgraduate program at your university.

I came into acquaintance with Xiaoyu in September 2022 in my class, during which he fully showed his intelligence and concentration in class, behaving in a friendly and constructive manner in inner-group cooperation. In the process of interacting with him, both inside and outside the classroom, I have come into acquaintance with his solid theoretical foundation and problem-solving ability. He demonstrated his systematic understanding of the parts like computer system architecture, data representation, algorithm principles, programming languages, equipping himself with a fundamental cognitive framework of computer science. He also actively participated in diverse algorithm design and programming practices involving network programming, concurrent programming, and performance optimization, to cultivate his ability tailored to solve complex problems.

As I recalled, he took the initiative to participate in numerous programming practices, during which he wrote simple programs to analyze the machine-level representation of data and the process of instruction execution. He was able to use the GDB debugging tool to track the changes of program variables and understand the correspondence between high-level languages and machine instructions. Simultaneously, he was also adept at precisely reverse-analyzing the binary programs, and deeply studied the buffer overflow attacks and program linking mechanisms. Such things enhanced his understanding of the underlying principles of computer systems and refined his competencies in programming and system design, earning him a good score--86.

In a nutshell, it is without any reservation that I recommend this outstanding individual for admission to your esteemed program. I believe that he will continue to make positive contributions to this program, and the academic environment and lived experience in your university will be very valuable to his development.

If you need further information, please feel free to contact me.

Sincerely

Signature: